Task 4 - Use Macros Safely

C Programming Assignment - Rural Electricity Billing System

Safe macro examples

#define RUPEES_TO_PAISE(r) ((long long)(r) * 100LL)

#define SQUARE(x) ((x) * (x))

#define MAX_VALUE(a,b) ((a) > (b) ? (a) : (b))

Parentheses

Without parentheses, SQUARE(2 + 3) can expand to 2 + 3 * 2 + 3 and give 11 instead of 25. Parenthesize every argument and the complete expression.

Multiple evaluation

MAX_VALUE(i++, j++) may evaluate an argument more than once. Avoid side effects in macro arguments and prefer functions when arguments may change state.

Expansion and comparison

Run gcc -E billing.c -o billing.i to inspect macro expansion. Macros lack type checking. static inline functions are safer and compilers often optimize them effectively.

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.